CISS245: Advanced Programming
Quiz q6901

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Q1. The following program

```
#include <iostream>

class Particle
{
public:
    Particle(int id,
             double position_x, double position_y,
             double velocity_x, double velocity_y)
        : id_(id),
          position_x_(position_x), position_y_(position_y),
          velocity_x_(velocity_x), velocity_y_(velocity_y)
    {}
    int id() const
    {
        return id_;
    }
    double position_x() const
    {
        return  position_x_;
    }
    double position_y() const
    {
        return  position_y_;
    }
    double velocity_x() const
    {
        return  velocity_x_;
    }
    double velocity_y() const
    {
        return  velocity_y_;
    }
private:
    int id_;
    double position_x_, position_y_;
    double velocity_x_, velocity_y_;
};

std::ostream & operator<<(std::ostream & cout, const Particle & particle)
{
```

```
    cout << "<Particle "
         << "id:" << particle.id() << ", "
         << "position:<" << particle.position_x() << ", "
         << particle.position_y() << ">, "
         << "velocity:<" << particle.velocity_x() << ", "
         << particle.velocity_y() << ">>";
    return cout;
}

int main()
{
    Particle particle(42, 0, 1, 2, 3); // particle at <0, 1> traveling with
                                       // velocity <2, 3>
    std::cout << particle << '\n';
    return 0;
}
```

has the following output:

```
<Particle id:42, position:<0, 1>, velocity:<2, 3>>
```

Rewrite the code using the given `vec2d` class.
Answer:

```
#include <iostream>

  class vec2d
  {
    public:
    vec2d(double x, double y)
    : x_(x), y_(y)
    {}
    double operator[](int i) const
    {
      return (i == 0 ? x_ : y_);
    }

    // other methods not shown
    private:
    double x_, y_;
  };

  std::ostream & operator<<(std::ostream & cout, const vec2d & v)
  {
    cout << '<' << v[0] << ", " << v[1] << '>';
    return cout;
  }

  class Particle
```

```
{
  public:
  Particle(int id, double position_x, double position_y,
  double velocity_x, double velocity_y)
  : id_(id), position_(position_x, position_y),
  velocity_(velocity_x, velocity_y)
  {}

  int id() const
  {
    return id_;
  }
  const vec2d& position() const
  {
    return position_;
  }

  const vec2d& velocity() const
  {
    return velocity_;
  }
  private:
  int id_;
  vec2d position_;
  vec2d velocity_;
};

std::ostream & operator<<(std::ostream & cout, const Particle & particle)
{
  // CLEAN UP THE CODE ACCORDINGLY.
  cout << "<Particle "
  << "id:" << particle.id() << ", "
  << "position:" << particle.position() << ", "
  << "velocity:" << particle.velocity() << ">";

  return cout;
}

int main()
{
  Particle particle(42, 0, 1, 2, 3); // particle at <0, 1> traveling with
  // velocity <2, 3>
  std::cout << particle << '\n';
  return 0;
}
```

Instructions

In the file `thispreamble.tex` look for

```
\renewcommand\AUTHOR{}
```

and enter your email address:

```
\renewcommand\AUTHOR{jdoe5@cougars.ccis.edu}
```

(This is not really necessary since alex will change that for you when you execute `make`.) In your bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

Enter your answers in `main.tex`. In the bash shell, execute "`make`" to recompile `main.pdf`. Execute "`make v`" to view `main.pdf`.

For each question, you'll see boxes for you to fill. For small boxes, if you see

```
1 + 1 = \answerbox{}.
```

you do this:

```
1 + 1 = \answerbox{2}.
```

`answerbox` will also appear in "true/false" and "multiple-choice" questions.

For longer answers that need typewriter font, if you see

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
\end{answercode}
```

you do this:

```
Write a C++ statement that declares an integer variable name x.
\begin{answercode}
int x;
\end{answercode}
```

`answercode` will appear in questions asking for code, algorithm, and program output. In this case, indentation and spacing is significant. For program output, I do look at spaces and newlines.

For long answers (not in typewriter font) if you see

```
What is the color of the sky?
\begin{answerlong}
\end{answerlong}
```

you can write

```
What is the color of the sky?
\begin{answerlong}
The color of the sky is blue.
\end{answerlong}
```

A question that begins with "T or F or M" requires you to identify whether it is true or false, or meaningless. "Meaningless" means something's wrong with the question and it is not well-defined. Something like "$1 + 2 = 4$" is either true or false (of course it's false). Something like "$1+_2 = 4$?" does not make sense.

When writing results of computations, make sure it's simplified. For instance write 2 instead of $1 + 1$.

Higher level classes.

For students beyond 245: You can put LaTeX commands in `answerlong`.

More examples of meaningless statements: Questions such as "Is $42 = 1+_2$ true or false?" or "Is $42 = \{2\}^{\{3\}}$ true or false?" does not make sense. "Is $P(42) = \{42\}$ true or false?" is meaningless because $P(X)$ is only defined if X is a set. For "Is $1 + 2 + 3$ true or false?", "$1 + 2 + 3$" is well–defined but as a "numerical expression", not as a "proposition", i.e., it cannot be true or false. Therefore "Is $1 + 2 + 3$ true or false?" is also not a well-defined question.

More examples of simplification: When you write down sets, if the answer is $\{1\}$, do not write $\{1, 1\}$. And when the values can be ordered, write the elements of the set in ascending order. When writing polynomials, begin with the highest degree term.

When writing a counterexample, always write the simplest.